Beyond Prediction: AI-Driven Energy Portfolio Decisions Under Uncertainty

From Machine Learning to Risk-Aware Capital Allocation

Alan Mössinger · SIADS 699 Capstone, University of Michigan · Group 13 (a solo project) · 2026

The decision

An energy company holds interconnection-queue requests and producing oil and gas leases. It must decide, request by request, whether to fund, hold or exit, and, across the whole book, how to split one budget. Both decisions are built from public records and scored on what happened. Four questions were set first: can machine learning **predict** better than transparent baselines, how do forecast error and correlated prices change **risk**, does a risk-aware book **allocate** better, and what **explains** the choice.

Part I — which queue requests get built

13,366 requests in CAISO, ERCOT and PJM; tested once on the 2,268 that entered in 2017–2019. ERCOT builds most often: 18.4% of entrants operate within five years, against 4.8% in CAISO.

ERCOT requests reach operation most often; CAISO's least

Aalen–Johansen cumulative incidence of operation (left) and withdrawal (right) by region; bands = 95% bootstrap (500 resamples); censored at 2024-12-31; ERCOT's withdrawal timing before Aug 2018 is distorted by a bulk date (section 2)

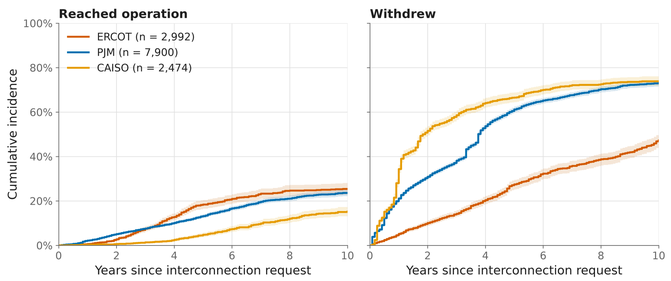


Figure: Alan Mössinger. Data: LBNL Queued Up: 2025 Edition (Rand et al., 2025). Cohort: Track A requests (CAISO, ERCOT, PJM) entering 2000–2024

At entry the model beats cohort averages (AUC 0.86 against 0.73) but ties a careful lookup at utility scale. **Two years in, with the agreement known, it earns its keep:**

Two years in, the model's book holds more operations

Books kept at the +2-year landmark by each chooser (no region above half), realised operations and MW within 3 years; 95% intervals from 200 resamples of the pool

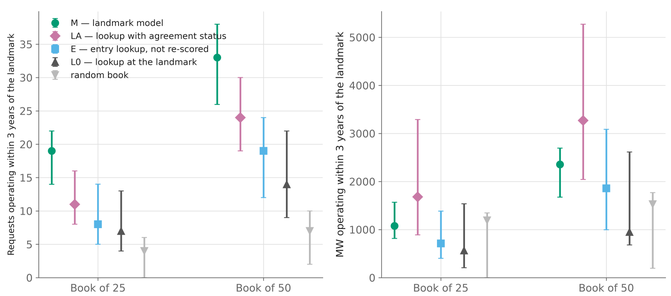


Figure: Alan Mössinger. Data: LBNL Queued Up: 2025 Edition (Rand et al., 2025). Cohort: Track A test entrants 2017–2019, ≥ 20 MW, in the queue two years after the request (n = 1,132)

A book of 25 chosen by the landmark model held 19 requests that reached operation, against 11 for the lookup. Every cohort since 2017 completes below the model's level, so books are sized to the newest cohort.

The work in one picture

One budget, four asset classes, five steps from data to decision

Part I's queue models give each power request its chance of operating by year; Part II values every candidate, draws them together and allocates one budget.

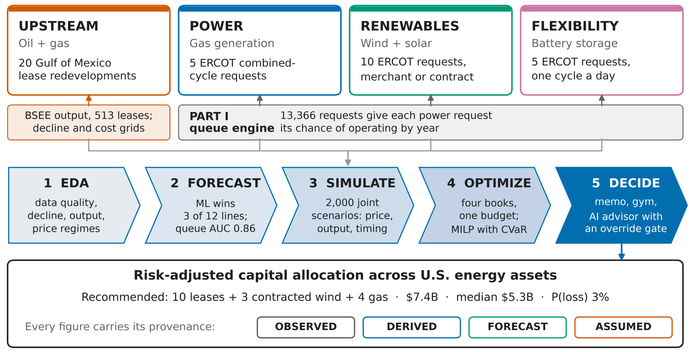


Figure: Alan Mössinger. Data: LBNL Queued Up: 2025 Edition (Rand et al., 2025). BSEE OGOR-A production (data.bsee.gov). EIA-923 via EIA API v2 (electricity/facility-fuel). EIA WTI and Henry Hub monthly spot prices. ERCOT DAM/RTM settlement point prices (gridstatus pull, 2024; ERCOT yearly DAM files, 2018–2023). Cohort: the forty candidates of Part II and the queue requests of Part I

Part II — one budget across twenty leases and twenty ERCOT requests

2,000 joint price scenarios drawn from observed prices, Winter Storm Uri included; Part I's models time every power request. Under the registered cost grid the leases are worth $14.6B and carry the book. **Priced as an investment committee would** — $40k per incremental flowing barrel a day and an 8% real rate, added after the registered results and labelled post-hoc — they fall to $7.0B on four times the capital, and five years of $42 oil take them to $4.25B.

A committee's grid shrinks the lease book and exposes a tail

Left: mean and median (5–95% bar) of the default book and book D under the registered and committee grids. Right: the twenty leases on the flat path under three oil paths. Post-hoc, labelled; contracted case.

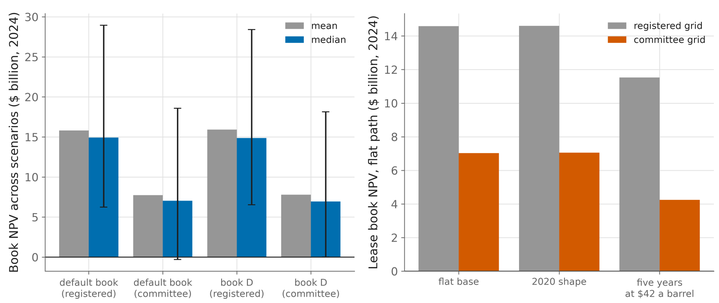


Figure: Alan Mössinger. Data: BSEE OGOR-A production (data.bsee.gov). EIA WTI and Henry Hub monthly spot prices. Provenance: ASSUMED. Cohort: the forty candidates of notebook 07; scenarios of notebook 08

Under the registered grid the risk-aware book earns $15.9B against $16.1B for NPV ranking and none of the four books' held-out tail is a loss, so two of the four allocation criteria pass and two fail. Under the committee grid the risk-aware book cuts the worst-5% loss from $1.3B to $0.3B; on the registered metric NPV ranking still wins by $888M.

The decision gym: rules against instinct, on outcomes

The registered rule beats instinct by $236M a situation

16 past situations; the advisor +228M in 16 recorded situations. Top: queue situations as of their date (realized value = modeled value per MW × realized operation − cost to the gate, a DERIVED proxy); in 6 of them none of the fifteen drawn requests operated within the window, so the best book is empty. Bottom: the lease regimes, scored on the realized price path. The tick is the best book with hindsight.

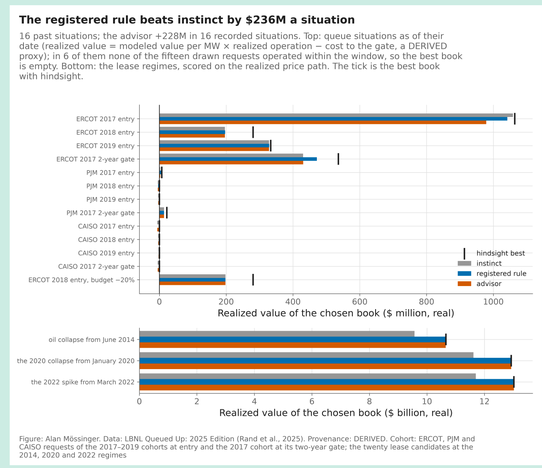


Figure: Alan Mössinger. Data: LBNL Queued Up: 2025 Edition (Rand et al., 2025). Provenance: DERIVED. Cohort: ERCOT, PJM and CAISO requests of the 2017–2019 cohorts at entry and the 2017 cohort at its two-year gate; the twenty lease candidates at the 2014, 2020 and 2022 regimes

Sixteen past situations rebuilt as of their date. The registered rules beat instinct by $236M a situation (regret against hindsight $18M against $253M), almost all of it in the three oil-price regimes.

The AI inside the decision

The advisor sees only result tables; humans override through a gate

Grey: data and records. Blue: the prompts and the check. Orange: the model and the gym. Green: the human and the gate. Arrows are the flow of one decision.

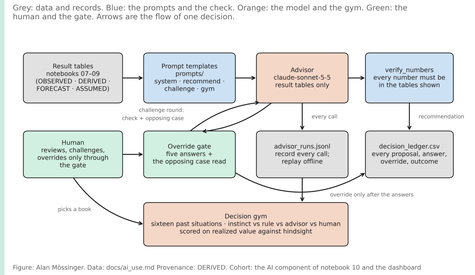


Figure: Alan Mössinger. Data: docs/ai_use.md Provenance: DERIVED. Cohort: the AI component of notebook 10 and the dashboard

An LLM advisor reads only the result tables, must match every number it writes to them, asks the human five questions and argues the opposing book, and can be overridden only after the answers. Run live on 96 recorded calls: 94% of its numbers are in its tables, its book overlaps the optimizer's by 0.89, and it beat instinct by $228M a situation.

Recommendation

Fund the risk-aware book: ten lease redevelopments, three wind requests under 15-year contracts and four gas requests, $7.4B of a $7.81B budget. It expects $5.8B (median $5.3B), loses money in 3% of scenarios, and its worst 5% averages a $0.3B loss. **The alternative** for a committee with more appetite for risk: eighteen leases and four gas requests, $7.7B expected with a $1.3B worst-5% loss. No solar or storage at 2024 prices.

Prediction is necessary and not sufficient: the decision is made by the assumptions, the timing and the tail. Data: LBNL Queued Up 2025 (Rand et al., 2025); BSEE OGOR-A; EIA-923 and EIA spot prices; ERCOT settlement-point prices; EPA eGRID; NREL ATB. Author's analysis.